Mahbod Nouri

ML Researcher
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github.com/mahbodnr

PhD researcher in Computational Neuroscience specializing in Deep Learning, Brain-inspired AI, and Computer Vision. Interested in optimizing neural networks for better efficiency and performance.

Education

PhD in Computational Neuroscience — Apr. 2023 – Present
University of Bremen — Bremen, Germany
- Focus: Efficient and Bio-plausible Deep Neural Networks

MSc in Cognitive Science — Sep. 2021 – Sep. 2022
University of Edinburgh — Edinburgh, United Kingdom
- Focus: Neuroscience and Artificial Intelligence
- Thesis: Bio-plausible Modeling of Associative Learning

BSc in Mechanical Engineering — Sep. 2016 – Sep. 2020
University of Tehran — Tehran, Iran
- Thesis: Deep Learning for Medical Image Segmentation

Selected Publications

Interleaving cortex-analog mixing improves deep non-negative matrix factorization networks
Mahbod Nouri, David Rotermund, Alberto Garcia-Ortiz, Klaus R. Pawelzik (2025)
Frontiers in Computational Neuroscience. doi: 10.3389/fncom.2025.1692418

BioLCNet: Reward-modulated Locally Connected Spiking Neural Networks
Hafez Ghaemi*, Erfan Mirzaei*, Mahbod Nouri*, Saeed Reza Kheradpisheh (2022).
Lecture Notes in Computer Science. doi: 10.1007/978-3-031-25891-6_42

Toward Real-World BCI: CCSPNet, A Compact Subject-Independent Motor Imagery Framework
Mahbod Nouri*, F. Moradi*, H. Ghaemi*, A. M. Nasrabadi (*Equal Contribution)(2022).
Digital Signal Processing. doi: 10.1016/j.dsp.2022.103816

Selected Conferences and Presentations

(Poster) Bernstein Conference in Computational Neuroscience — Sep. 2024
Frankfurt am Main, Germany

(Poster) Nordic Probabilistic AI School (ProbAI) — June. 2024
Copenhagen, Denmark

Cortical Prostheses – Interdisciplinary Research towards Artificial Vision for the Blind — Sep. 2023
Hanse-Wissenschaftskolleg, Delmenhorst, Germany

(Oral) 2nd Advanced Course and Symposium on Artificial Intelligence and Neuroscience — Sep. 2022
Certosa di Pontignano, Castelnuovo Berardenga (Siena), Tuscany, Italy

Internship

AI Research Intern Summer 2020
Artificial Intelligence in Mechanical Engineering (AIME) Lab University of Tehran

- Developed a COVID-19 diagnosis model from crowd-sourced cough audio data

Selected Projects

COVID-19 Lung Segmentation 2020
Collected and preprocessed CT scan data from multiple sources and trained a deep learning model to segment COVID-19 infections in the lungs.

Real-Time Mask Detection System 2021
Developed and deployed a face mask detection model that provided real-time visual feedback through a live camera interface.

LLMafia 2025
Created an interactive playground where multiple large language models play the Mafia game, featuring a user-friendly interface for observing and analyzing their behavior.

DubGAN 2025
An end-to-end GAN framework for multilingual dubbing that preserves speaker and acoustic features.

Latent Space 2026
An ARG-style bimonthly puzzle for fellow geeks. mahbod.me/latent-space

Volunteer Experience

Lecturer since Jan. 2021
Python Programming Lecturer at Kaaryar College, a social start-up providing programming education

Organizing Committee Sep. 2023
Organizing the *Machine Learning and its Applications* summer school at the University of Bremen

Teaching Experience

Tutor since 2023
Tutored Master's courses in *Mathematics and Computational Neuroscience* courses at the University of Bremen.

Skills

Languages: English (C1), German (A2), Farsi/Persian (Native)
Programming: Python (PyTorch, TensorFlow, Keras, Scikit-learn, Flask), Julia, C++, R
Age of Empires 2: I'm really good at it.

Honors and Awards

Fifth National Brain-Computer Interface (BCI) Competition June 2022
Third Rank in the National Brain-Computer Interface (BCI) Competition
Focusing on Auditory Attention held by National Brain Mapping Laboratory (NBML)

Oxford Hackathon Feb. 2022
Winner of the Sponsor Challenge

HackTheBurgh VIII Hackathon Feb. 2022
Second Place

LBSV Football League 2024
3rd League Champions with the University of Bremen Team